Mathematics - Graduate Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the value of the expression 2,816 x 7?

   (a) 19,812
   (b) 20,112
   (c) 19,612
   (d) 14,572
   (e) 18,712

2. Suppose $x-3$ and $y+3$ are multiples of 7. What is the smallest positive integer, $n,$ for which x^2+xy+y^2+n is a multiple of 7?

   (a) 9
   (b) 8
   (c) 5
   (d) 6
   (e) 3

3. If four points are picked independently at random inside the triangle ABC, what is the probability that no one of them lies inside the triangle formed by the other three?

   (a) 0.1250
   (b) 0.6667
   (c) 0.4444
   (d) 0.8000
   (e) 0.5000

4. What is the greatest common factor of 36 and 90?

   (a) 30
   (b) 12
   (c) 45
   (d) 15
   (e) 6

5. Find the last 3 digits of 2003^(2002^2001).

   (a) 806
   (b) 001

(c) 615

(d) 404

(e) 923

True / False (10 marks)

Answer True or False

6. True or False: The solution to the equation v - 26 = 68 is v = 84.
7. True or False: The Binormal vector for the given vector function r(t) is [0.7, 0.0, 0.7].
8. True or False: If g(x) = f(-x) and g(3) = 2, then f(-2) must equal 3.
9. True or False: If a sum is invested at a continuous annual return rate of 7%, it will double in approximately 9.90 years.
10. True or False: Values such as 0.0042 and 0.005 are also acceptable choices for x within the spe

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the curve defined by the equation

$$\theta = \frac{\pi}{-}a$$

(A) Line (B) Circle (C) Parabola (D) Ellipse (E) Hyperbola Enter the letter of the correct option.

12. Convert 813_9 to base 3.

End of Paper